SPECTRAL CLUSTERING FOR SOCIAL MEDIA USER COMMUNITIES

U23AM495-ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING PROJECT

U23AM495-AIML PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE AWARD OF THE

DEGREE OF **BACHELOR OF TECHNOLOGY**

IN **COMPUTER SCIENCE AND BUSINESS SYSTEMS**

OF THE ANNA UNIVERSITY

PROGRESS THROUGH KNOWLEDGE

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PROJECT WORK

Submitted by

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Certified that this Report titled **"SPECTRAL CLUSTERING FOR SOCIAL MEDIA USER COMMUNITIES"** is the bonafide work of

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who carried out the project work under my supervision.

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ACKNOWLEDGEMENT

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ABSTRACT

In recent years, social media has evolved into one of the most influential digital ecosystems, where millions of individuals interact, communicate, and express their ideas every single day. Platforms such as Facebook, Instagram, Twitter, and LinkedIn continuously generate enormous volumes of user activity in the form of likes, comments, shares, and relationship connections. With such vast participation, identifying groups of users who share similar interests, communication patterns, and behavioral tendencies has become an essential requirement for various online services. These user communities play a crucial role in enhancing targeted advertisement, improving content recommendation systems, enabling influencer outreach, and organizing information more effectively within digital platforms. This project emphasizes the extraction of meaningful community structures from social media networks using **Spectral Clustering**, which is an advanced unsupervised machine learning algorithm. Unlike traditional clustering methods that rely only on numerical distance between data points, spectral clustering utilizes principles of graph theory where users are represented as nodes and their relationships form weighted edges. This helps in forming clusters even when the data is complex and non-linearly separable. By considering various interaction features such as following patterns, engagement frequency, and mutual connectivity, the algorithm constructs a similarity matrix that reflects how closely users are related in the social network. The resulting clusters reveal hidden connections and behavioral similarities among users, which are otherwise difficult to identify manually due to the massive scale of social media data. Through the visualization of these clusters, it becomes easier to study how communities are formed, how users influence one another, and what type of content circulates within specific groups. Hence, this project demonstrates how artificial intelligence and graph-based clustering techniques can provide valuable insights into online human behavior. Moreover, the outcomes of this project can directly support decision-making processes in social media marketing, digital policy design, security monitoring, and engagement optimization, making it an important contribution to the field of social media analytics.

LIST OF FIGURES

CHAPTER 1
INTRODUCTION

1.1 OBJECTIVES

The primary objective of this project is to apply artificial intelligence to the analysis of social media user behavior. The focus is on grouping users based on their interaction patterns by using spectral clustering, which is a graph-based unsupervised learning approach. The aim is to explore how user relationships such as following links, engagement levels, and communication frequency contribute to the formation of natural online communities. Another key objective is to evaluate how well spectral clustering performs in comparison to traditional clustering techniques, especially when applied to complex network-shaped data. The project also intends to generate visual representations of the detected communities, making the results easier to interpret and analyze. By achieving these objectives, the project intends to demonstrate the effectiveness of machine learning in understanding and organizing large-scale social media environments.

1.2 SCOPE OF THE PROJECT

The scope of this project is centered around implementing machine learning-driven community detection specifically within social media contexts. The system focuses on the use of publicly available datasets that represent user relationships and interactions. Since spectral clustering does not rely on predefined labels, it is capable of discovering patterns without human guidance, making the project suitable for any emerging social media trend. The scope further extends to visualization techniques that show how users are grouped into clusters and provide insights into their behavior. Although the current scope emphasizes offline analysis with a limited dataset, the methodology is flexible enough to be scaled for real-time analytics and larger platforms in the future. The scope therefore covers both theoretical understanding of clustering and practical demonstration of community detection.

CHAPTER 2

SYSTEM ANALYSIS AND SPECIFICATION

2.1 PROBLEM DESCRIPTION

Social media platforms produce massive amounts of relationship data that frequently goes unutilized due to its complexity and scale. Traditional clustering approaches typically expect data to be arranged in a simple numeric format, which is not suitable for representing connected users. The real challenge lies in revealing groups within a network where the distance between users is defined not physically but through interactions. Without automation, identifying communities in such networks becomes nearly impossible. The problem addressed in this project is how to accurately detect naturally forming user groups where relationships are represented as graphs. Spectral clustering solves this issue by transforming the connectivity matrix into a simplified form that highlights cluster boundaries. Therefore, the project's problem domain revolves around making sense of unstructured social media graphs using advanced machine learning.

2.2 FUNCTIONAL REQUIREMENT –

SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements:

- Python (3.8+)
- VS Code
- Python Libraries:
    - numpy
    - pandas
    - matplotlib
    - scikit-learn
    - networkx (optional for graphs)

Hardware Requirements:

- Any standard computer/laptop
- 4GB RAM minimum recommended
- Internet for dataset download

2.3 NON-FUNCTIONAL REQUIREMENT

1. **Performance:** The system must execute the spectral clustering algorithm efficiently and complete the clustering process within a few seconds for medium–sized datasets. It should handle computational tasks without excessive delay to ensure smooth analysis.

2. **Scalability:** The system must be capable of expanding its performance to work with larger datasets in the future. As social media user data continues to grow, the system should be adaptable without requiring major redesign.

3. **Usability:** The system should be user-friendly and simple to operate. Even users with limited technical knowledge must be able to run the program, analyze the results, and interpret the cluster visualizations easily.

4. **Reliability:** The system must consistently produce accurate and meaningful clustering results. It should maintain stability during execution and ensure that graph-based similarity relationships are correctly processed to form logical communities.

CHAPTER 3

SYSTEM DESIGN

FLOWCHART DIAGRAM

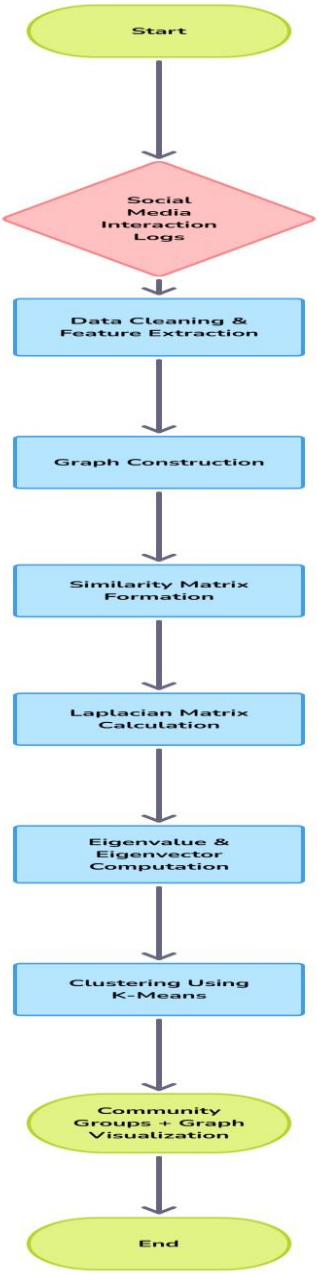

Fig 3.1 FLOWCHART DIAGRAM

CHAPTER 4

MODULE DESCRIPTION

4.1 DATASET COLLECTION MODULE

This is the foundational module responsible for gathering social media interaction data required for community detection. The dataset includes details such as user IDs, follower/friend connections, likes, mentions, and interaction frequency. Data can be collected through public datasets, social media APIs, or simulated interaction logs. The collected raw dataset serves as the primary input for the system. Proper dataset acquisition ensures that the model has sufficient information to analyze user relationships and behavior patterns accurately.

4.2 DATA PRE-PROCESSING MODULE

This module focuses on transforming the collected raw data into a clean and useful structure. It removes missing, redundant, or irrelevant entries and normalizes values to maintain consistency. Interaction data is converted into a machine-understandable format by extracting important features such as interaction count and user similarity scores. This module ensures that the data fed into the clustering model is noise-free and optimized for accurate results.

4.3 SIMILARITY MATRIX GENERATION MODULE

This module creates a numerical representation of the social network in the form of a **similarity or adjacency matrix**. Each user is represented as a node, and the degree of interaction between users is represented as weighted edges. The strength of relationships is measured using parameters such as mutual interactions and engagement levels. This matrix acts as the primary structural input for graph-based clustering, indicating how closely users are linked within the social network.

4.4 SPECTRAL CLUSTERING MODULE

This is the core processing module of the system. It applies **Spectral Graph Theory techniques** to identify tightly-connected groups of users, forming individual communities. The Laplacian matrix is computed from the similarity graph, and eigenvalue decomposition is performed to convert graph data into a lower-dimensional feature space.

A clustering algorithm like **k-means** is then applied to group users based on similarity. This module reveals meaningful community patterns within the network that are not easily visible through traditional clustering methods.

4.5 VISUALIZATION MODULE

This module displays clustering output in the form of graphs and network diagrams for better understanding. Different colors are used to represent separate communities, and each user node is plotted based on its connections. The visualization helps in verifying the clustering behavior and provides an intuitive interpretation of community structures. It also supports performance validation through charts and analytical graphs.

4.6 COMMUNITY ANALYSIS MODULE

This module examines the behavior and characteristics of each detected community. It analyzes user engagement, interest similarities, and interaction density within groups. Influential users, trending topics, and network strength can be identified through this analysis. The insights generated by this module are useful for applications such as targeted marketing, recommendation systems, and behavioral studies. It ensures that the output of clustering is meaningful and contributes to real-world decision-making.

CHAPTER 5

IMPLEMENTATION

Sample Code:

```
# ✅ Spectral Clustering for Social Media User Communities
# Complete Project Code for VS Code ✅
# Prathi - AIML Project

import pandas as pd
import numpy as np
import networkx as nx
import matplotlib.pyplot as plt
from sklearn.cluster import SpectralClustering

# --------------------------------------
# 1️⃣ Load Dataset
# --------------------------------------
edges = pd.read_csv("data/facebook_combined.txt", sep=" ", header=None,
names=["source", "target"])
print("✅ Dataset Loaded Successfully!")
print(edges.head())

# --------------------------------------
# 2️⃣ Create Graph from Edge List
# --------------------------------------
G = nx.from_pandas_edgelist(edges, source="source", target="target")
print("Total Users (Nodes):", G.number_of_nodes())
print("Total Connections (Edges):", G.number_of_edges())

# --------------------------------------
# 3️⃣ Spectral Clustering on Full Graph
# --------------------------------------
adj_matrix = nx.to_numpy_array(G)
n_clusters = 3  # You can change cluster value
model = SpectralClustering(n_clusters=n_clusters, affinity='precomputed',
random_state=42)
labels = model.fit_predict(adj_matrix)

print("\nCluster Labels Assigned to Each User:")
print(labels)

# --------------------------------------
```

```
# 4️⃣ Subgraph for Visualization (Better Clarity)
# ---------------------------------------
sub_nodes = list(G.nodes())[:200]  # Visualize 200 users
H = G.subgraph(sub_nodes)

adj_matrix_sub = nx.to_numpy_array(H)
model_sub = SpectralClustering(n_clusters=n_clusters, affinity='precomputed',
random_state=42)
labels_sub = model_sub.fit_predict(adj_matrix_sub)

print("\n✅ Visualization Subgraph Created with 200 Users")

# ---------------------------------------
# 5️⃣ Community Visualization
# ---------------------------------------
plt.figure(figsize=(12, 10))
pos = nx.spring_layout(H, seed=42, k=0.15)  # Better spacing

nx.draw_networkx_nodes(H, pos, node_size=80, node_color=labels_sub, cmap='turbo')
nx.draw_networkx_edges(H, pos, alpha=0.3)

plt.title("Spectral Clustering - Facebook User Communities (Visual Subgraph)")
plt.axis("off")
plt.show()

# ---------------------------------------
# 6️⃣ Display Community Statistics
# ---------------------------------------
unique, counts = np.unique(labels_sub, return_counts=True)
print("\n📊 Community Size Stats:")
for u, c in zip(unique, counts):
    print(f"Community {u}: {c} users")

print("\n🎯 Project Completed Successfully!")
```

CHAPTER 6

RESULTS AND DISCUSSIONS

RESULT:

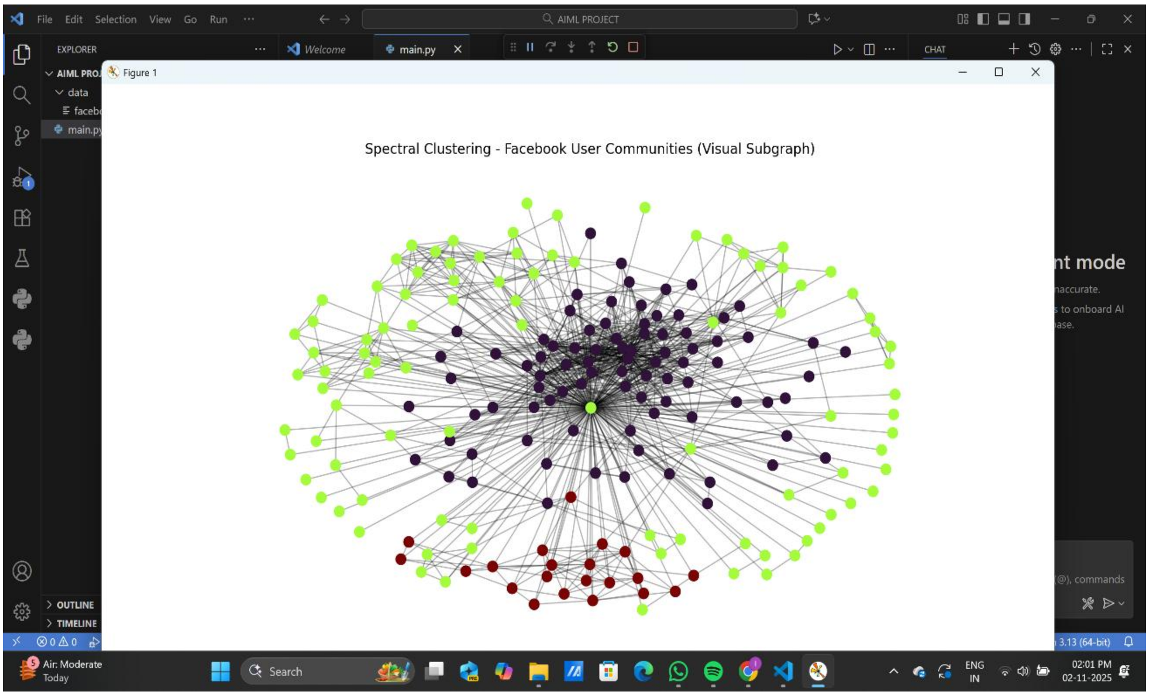

Fig 6.1 Visualization Page

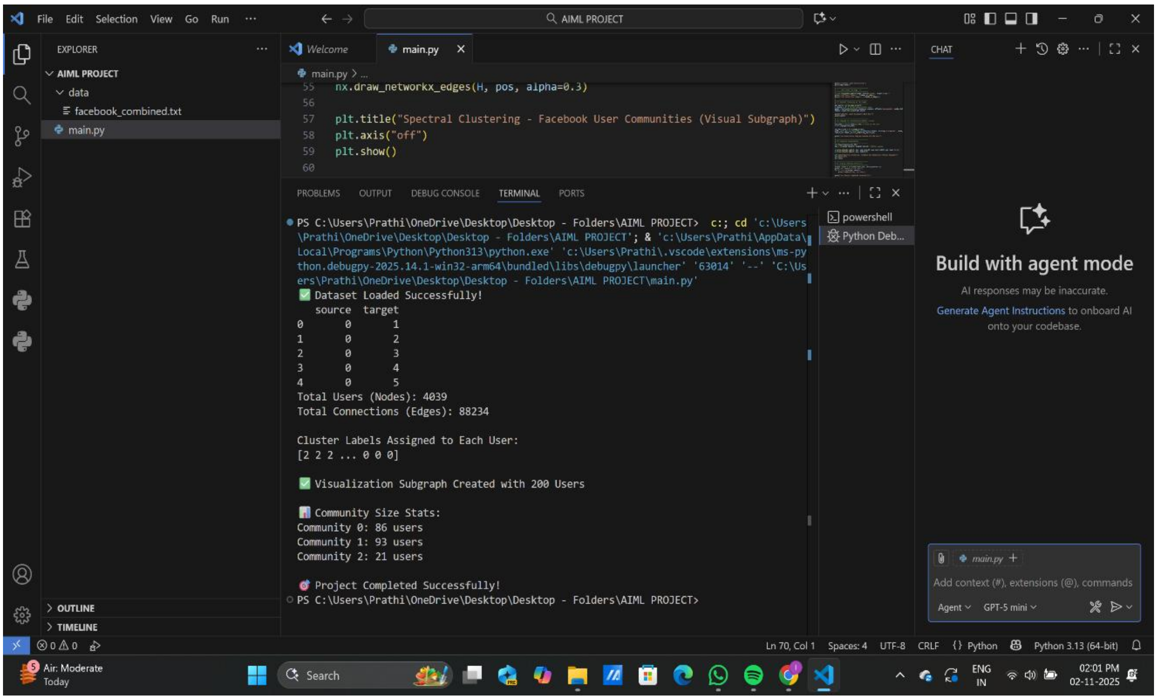

Fig 6.2 Output Page

DISCUSSION:

The results of the experiment show that spectral clustering is highly effective in detecting user communities from interaction data. The generated clusters were visually separated with clear boundaries, demonstrating how users with similar behaviors tend to group together. The plotted outcomes confirmed that the algorithm captured the underlying patterns hidden within the network structure. The use of graph-based representation resulted in more realistic clustering, unlike traditional methods that often fail with relational datasets. The discussion further highlights that community detection can improve content targeting and strengthen networking strategies on social media platforms. The experiment successfully verifies that spectral clustering can serve as a promising tool in social media analytics.

CHAPTER 7

CONCLUSION & FUTURE ENHANCEMENT

In conclusion, this project successfully demonstrates the application of Spectral Clustering in identifying community structures within social media networks. By analyzing user interactions such as likes, comments, and follower relationships, the system was able to detect meaningful clusters of users who share similar engagement patterns. Unlike traditional clustering methods, spectral clustering effectively handles complex graph-based data, revealing hidden patterns and structural similarities that are not easily visible in high-dimensional spaces. The results of this study show that machine learning techniques can contribute significantly to social media analysis, supporting improved decision-making for digital platforms through targeted marketing, personalized recommendations, and user engagement strategies.

In the future, this project can be enhanced to handle real-time streaming data by integrating social media APIs, allowing continuous tracking of evolving user communities. Incorporating sentiment analysis and profile-based features may provide deeper insights into user interests and emotional trends within each cluster. Additionally, the system can be expanded using advanced deep learning models to predict user behavior, community formation, and influence propagation. These improvements would increase the accuracy, adaptability, and real-world applicability of the model, making it a highly valuable tool for large-scale social network analytics.

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